



WineQuality

inteRstellar



Agenda

Description of the dataset

Simple models and meaningful
variables

Task 1: Regression

Task 2: Clustering

Conclusions





Introduction

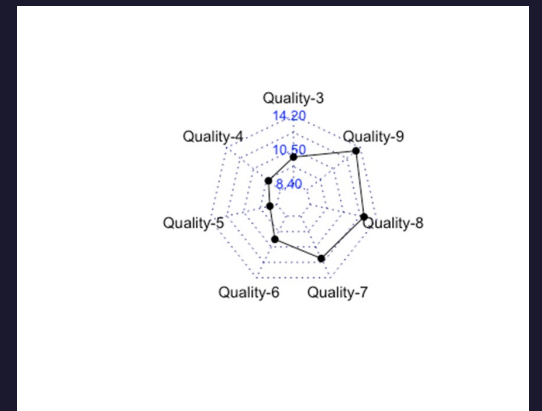
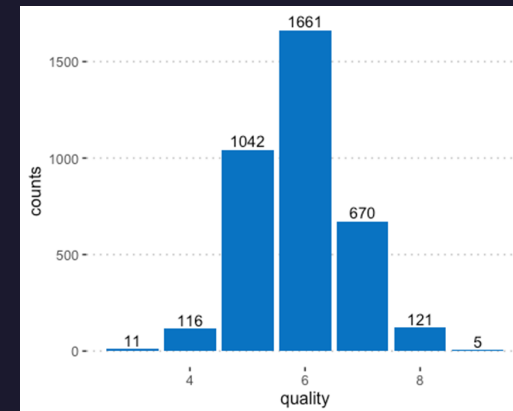
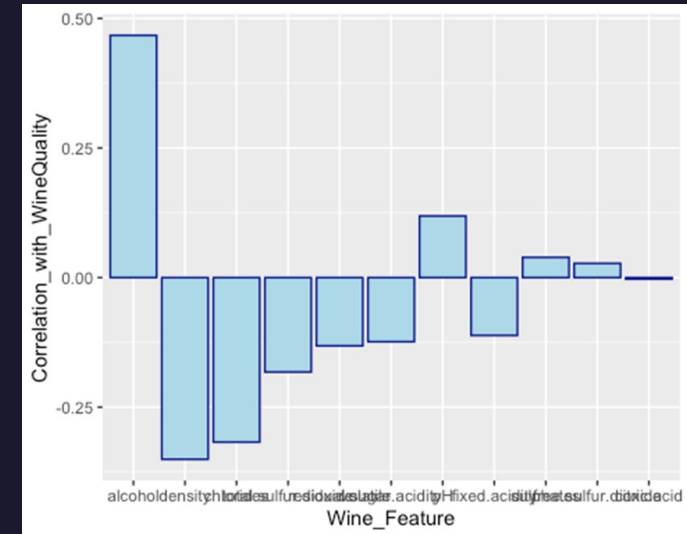
- This dataset contains information about the white variant of the Portuguese Vinho Verde. Wine samples are recorded on the rows.
- In a business perspective it is undoubtedly important to catch the correct quality of the 'in working wine' and consequently to sell it at the right price.

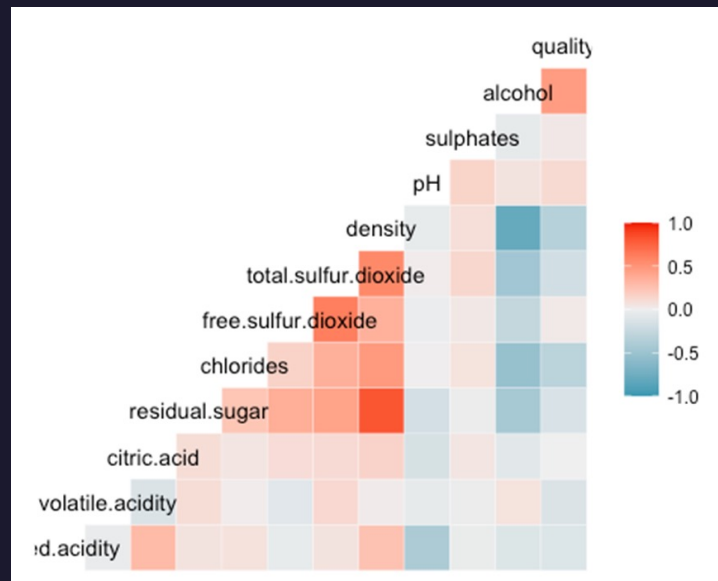
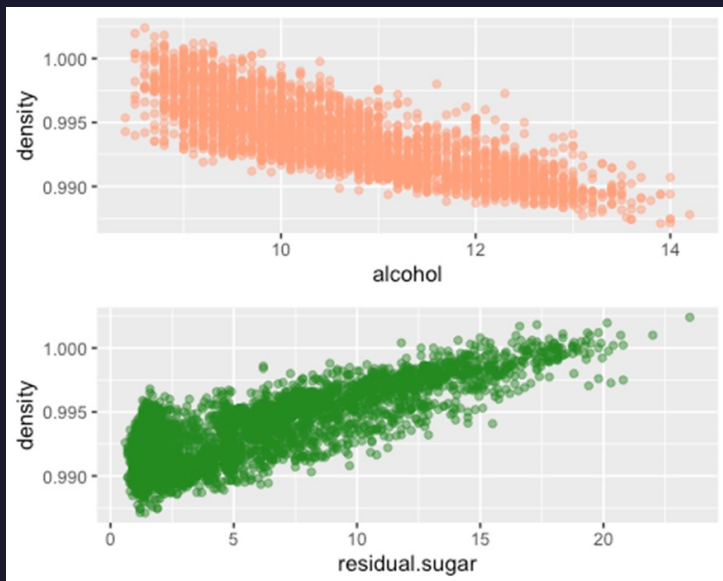
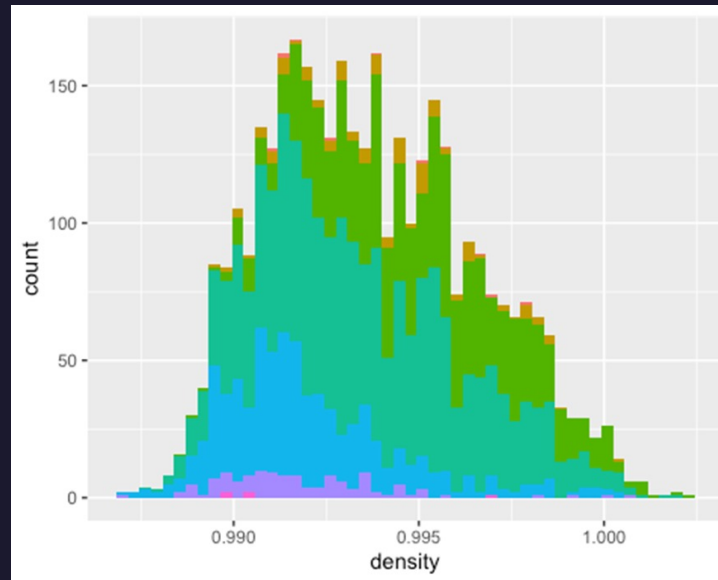
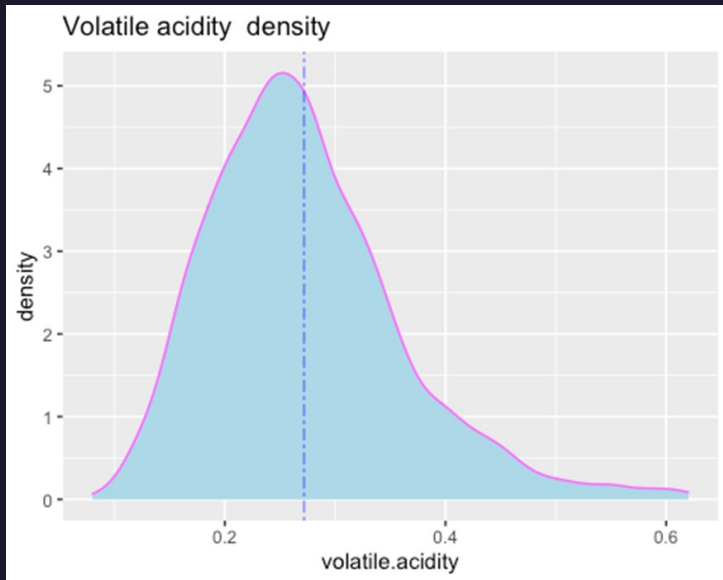
Task 1: Regression

To predict the wine quality we build a regression algorithm and compare the results through MSE.

General Overview

- 1) Clean the dataset: Nas, Duplicates, Outliers
- 2) What about the response variable 'quality'?





The predictors:

- Fixed acidity
- Volatile acidity
- Citric acid
- Residual sugar
- Chlorides
- Free sulfur dioxide
- Total sulfur dioxide
- Density
- Ph
- Sulphates
- Alcohol

Investigating the most meaningful variables and lower dimensional models

-Multiple Regression with all the predictors

	2.5 %	97.5 %
(Intercept)	1.427886e+02	2.499583e+02
fixed.acidity	7.421574e-02	1.802663e-01
volatile.acidity	-1.620138e+00	-1.036979e+00
citric.acid	-1.841167e-02	5.050329e-01
residual.sugar	6.750827e-02	1.076132e-01
chlorides	-7.798881e+00	-2.364676e+00
free.sulfur.dioxide	5.572170e-03	9.739541e-03
total.sulfur.dioxide	-1.245266e-03	5.287012e-04
density	-2.525051e+02	-1.438813e+02
pH	8.893596e-01	1.398506e+00
sulphates	4.896776e-01	9.657218e-01
alcohol	6.043171e-02	1.942223e-01

-Multiple Regression with Alcohol and Volatile Acidity

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.64079	0.11620	22.73	<2e-16 ***
alcohol	0.34662	0.01049	33.03	<2e-16 ***
volatile.acidity	-1.57673	0.14442	-10.92	<2e-16 ***

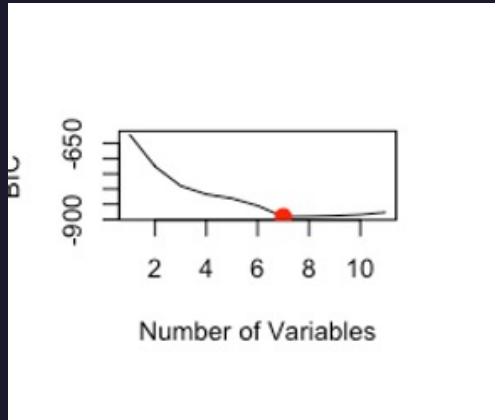
-Adding an interaction term

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.22742	0.37516	13.934	< 2e-16 ***
alcohol	0.10264	0.03525	2.912	0.00361 **
volatile.acidity	-10.31007	1.21377	-8.494	< 2e-16 ***
alcohol:volatile.acidity	0.82175	0.11341	7.246	5.22e-13 ***

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fixed.acidity	1	34.42	34.417	64.1840	1.515e-15 ***
volatile.acidity	1	45.02	45.021	83.9602	< 2.2e-16 ***
citric.acid	1	0.70	0.695	1.2966	0.254917
residual.sugar	1	36.46	36.461	67.9956	2.268e-16 ***
chlorides	1	231.71	231.713	432.1209	< 2.2e-16 ***
free.sulfur.dioxide	1	18.36	18.365	34.2487	5.282e-09 ***
total.sulfur.dioxide	1	36.35	36.346	67.7818	2.523e-16 ***
density	1	262.44	262.439	489.4213	< 2.2e-16 ***
pH	1	124.52	124.516	232.2098	< 2.2e-16 ***
sulphates	1	28.82	28.823	53.7514	2.794e-13 ***
alcohol	1	7.47	7.468	13.9263	0.000193 ***
Residuals	3614	1937.91	0.536		

-Gam

Linear Methods



Subset selection

Metric: BIC

	Df	Sum of Sq	RSS	AIC
<none>			1933.3	-905.33
+ total.sulfur.dioxide	1	0.086	1933.2	-903.45
- citric.acid	1	3.025	1936.3	-903.08
- alcohol	1	3.528	1936.8	-902.37
- chlorides	1	5.391	1938.7	-899.76
- fixed.acidity	1	12.912	1946.2	-889.23
- sulphates	1	24.233	1957.5	-873.46
- volatile.acidity	1	38.419	1971.7	-853.83
- density	1	41.188	1974.5	-850.01
- free.sulfur.dioxide	1	49.721	1983.0	-838.29
- residual.sugar	1	51.630	1984.9	-835.67
- pH	1	54.038	1987.3	-832.37

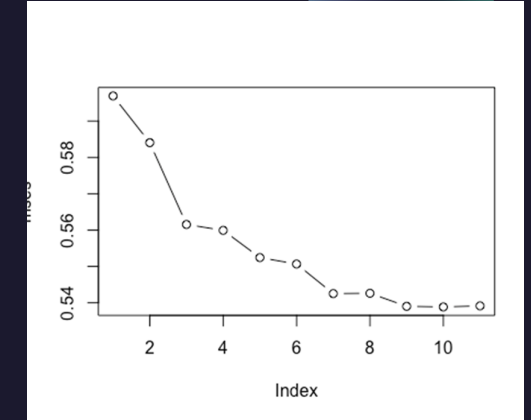
Stepwise Selection

Metric :AIC

	Df	Sum of Sq	RSS	AIC
<none>			1945.5	-846.88
+ chlorides	1	5.05	1940.5	-846.04
+ citric.acid	1	3.77	1941.8	-844.25
+ alcohol	1	3.72	1941.8	-844.18
+ total.sulfur.dioxide	1	0.02	1945.5	-838.99
- sulphates	1	35.25	1980.8	-805.97
- volatile.acidity	1	40.50	1986.0	-798.76
- free.sulfur.dioxide	1	50.53	1996.0	-785.07
- fixed.acidity	1	62.23	2007.8	-769.17
- pH	1	132.76	2078.3	-675.30
- residual.sugar	1	319.93	2265.4	-440.83
- density	1	658.66	2604.2	-61.96

Stepwise Selection

Metric : BIC



K-fold Cross Validation

Metric :MSE

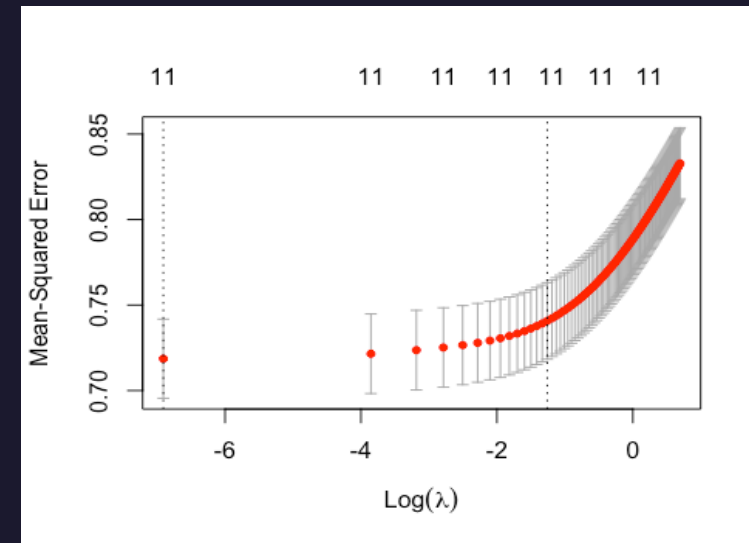
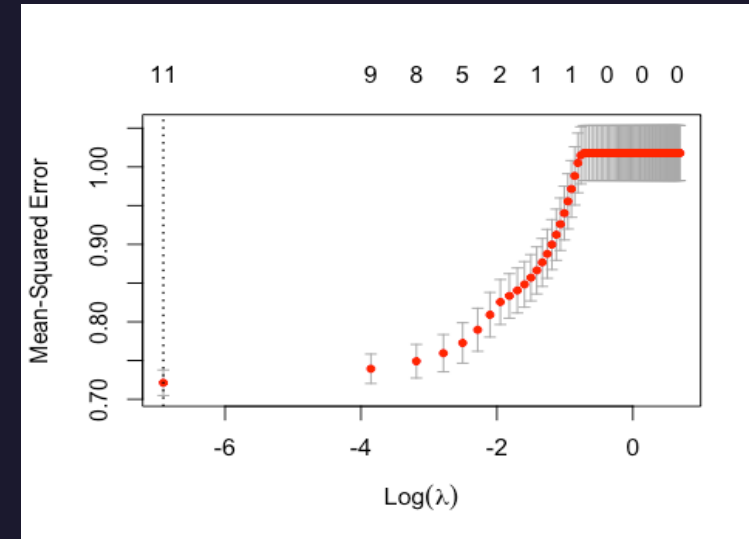
Shrinkage Methods

LASSO

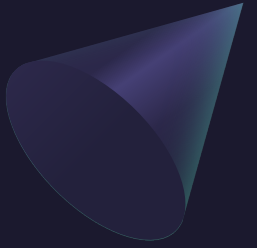
- First we fit model
- Then we perform cross validation on λ to catch the min MSE
- Lastly we fit the new model and perform the predictions with the resulting λ

RIDGE

- Same procedure with Ridge



Non-linear Methods



NATURAL SPLINES



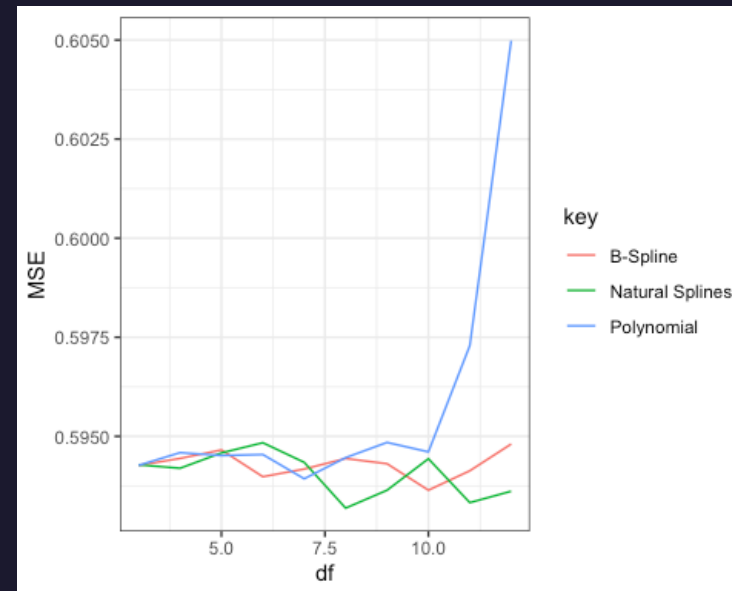
B-SPLINES

POLYNOMIALS

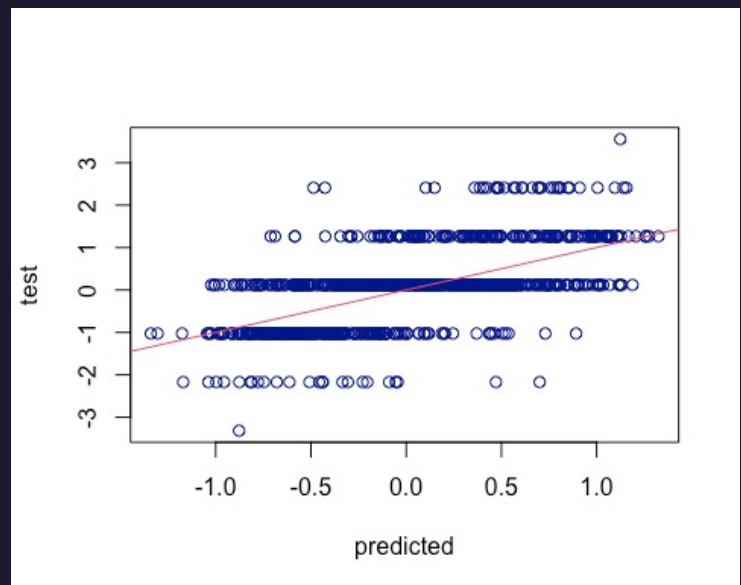
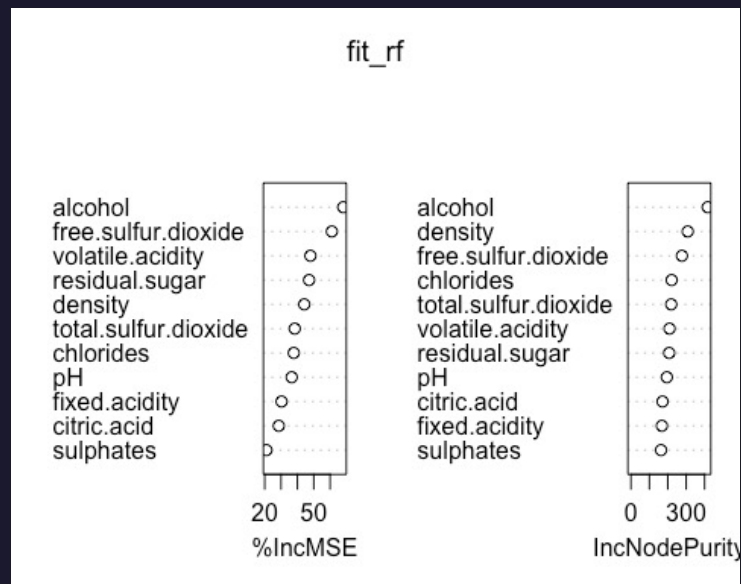
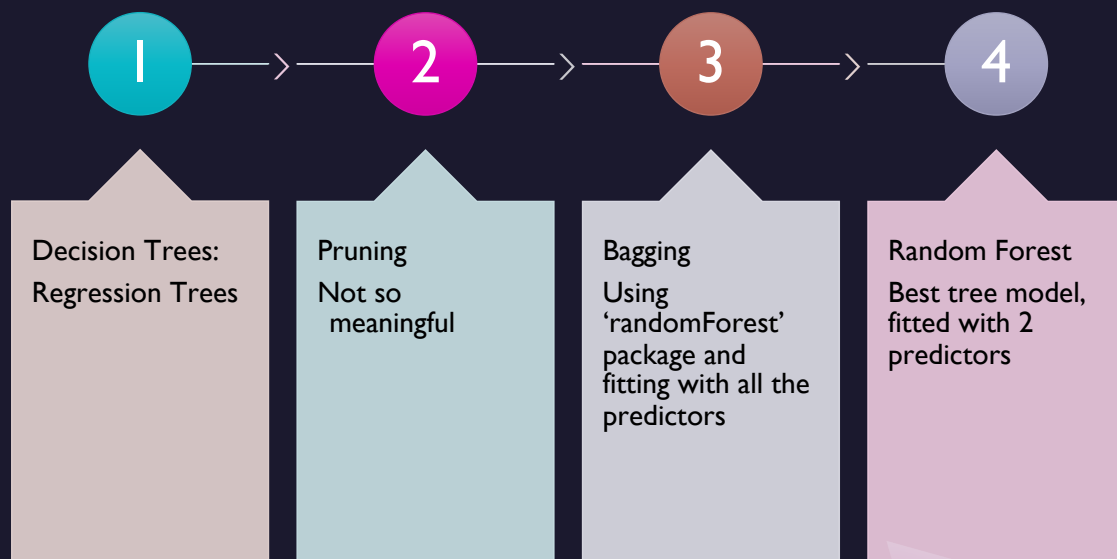


First we use 10-folds cross validation to find the degrees of freedom which minimize the MSE in the training set

Polynomial	Natural Splines	B-Spline
0.5942684	0.5942819	0.5942684



Trees





What about the results?

-Best Model: Multiple linear Regression fitted with 10 predictors (result of Cross Validation)

-Mse around 0.54

```
##                               10
## 0.6865987 0.8989521 0.8173199 0.5388563 0.8142569 0.6757247 0.7149075 0.71
22630
##
## 0.7157628 0.7028344 0.7116510 0.5962608 0.5938442
min(vector)
## [1] 0.5388563
```



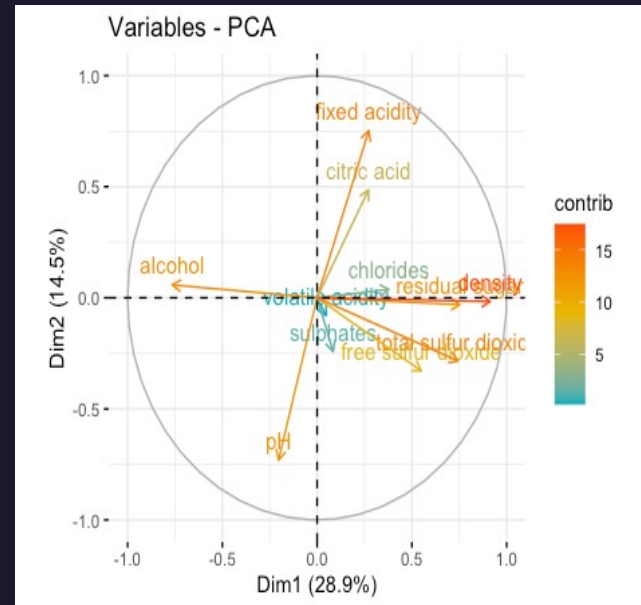
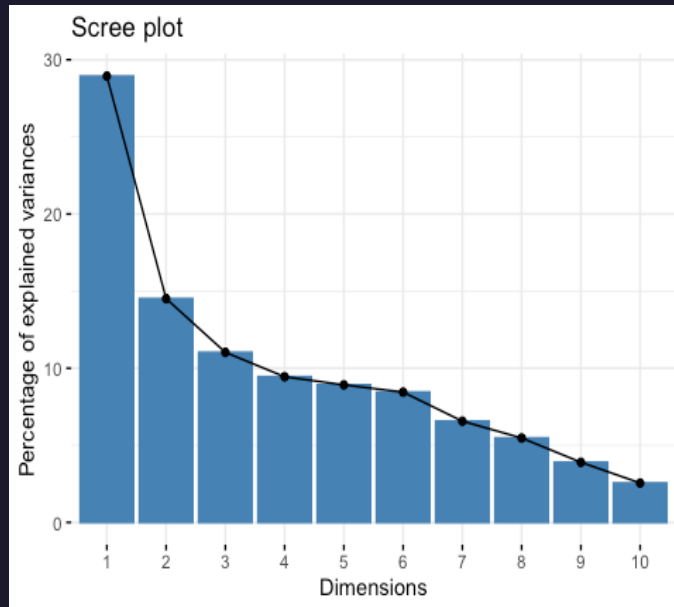

Task 2: Clustering

In a business perspective, to give a firm a full understanding of which chemical characteristics make a "good wine", we performed a clustering



PCA

- 1) We applied pca to our dataset and extracted the most meaningful 5 components
- 2) Analyzed the components to understand which initial predictors were more influential



	Dim.1	Dim.2	Dim.3	Dim.4
fixed acidity	0.27269588	0.75303655	0.06108969	0.003124276
volatile acidity	0.04776586	-0.08003547	-0.74034852	0.171504141
citric acid	0.27104607	0.48522968	0.50823689	0.145063472
residual sugar	0.75312133	-0.03097378	-0.22272688	-0.285498538
chlorides	0.37820238	0.03752709	-0.11959338	0.682356858
free sulfur dioxide	0.54982038	-0.32905066	0.34134241	-0.344151068
total sulfur dioxide	0.74368505	-0.28454944	0.13590691	-0.095729362
density	0.91360233	-0.01655750	-0.14190667	0.006318739
pH	-0.20343973	-0.73065204	0.15970583	0.136720814
sulphates	0.08464282	-0.24173161	0.38858427	0.507559422
alcohol	-0.76459508	0.05811486	0.08577522	-0.193667589
	Dim.5	Dim.6	Dim.7	Dim.8
fixed acidity	0.257588259	0.002740261	0.17936932	0.43755274
volatile acidity	0.380214294	0.417458359	-0.24232229	0.08384854

K-means clustering

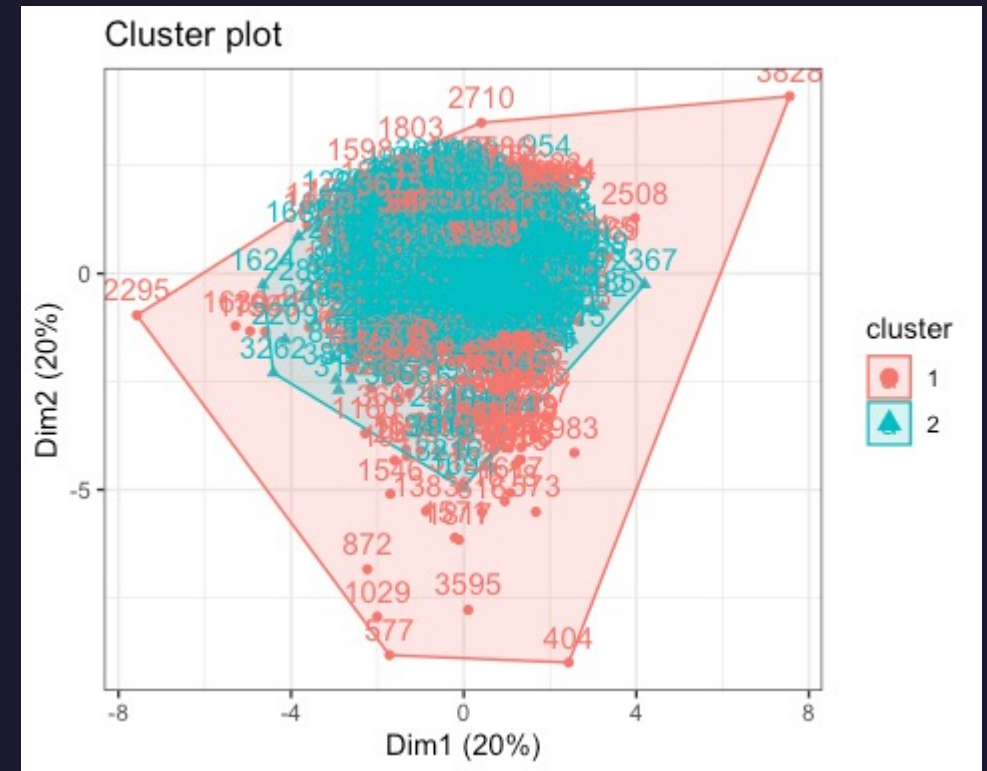
- 1) Applied K-means algorithm on our pca_data using 2 clusters as result from Silhouette and Elbow Method
- 2) Extracted the most meaningful components

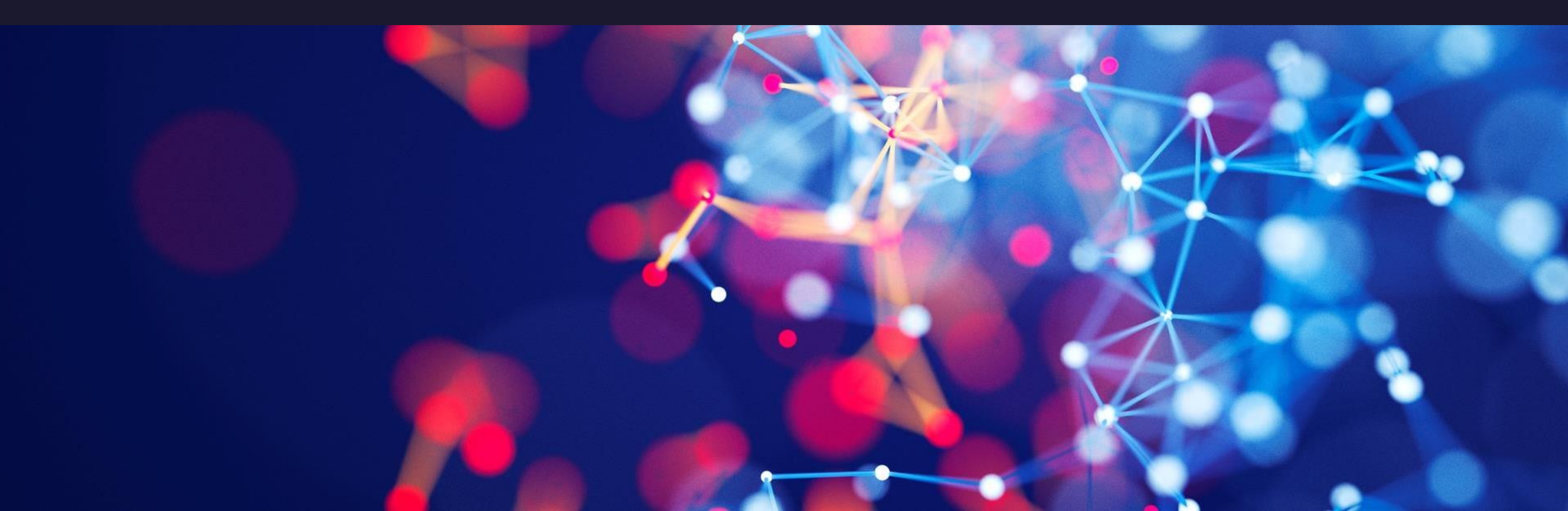
Cluster means:

	pc.comp1	pc.comp2	pc.comp3	pc.comp4	pc.comp5
1	-1.921356	0.11281346	0.03478809	0.04874152	0.03162125
2	1.153590	-0.06773366	-0.02088691	-0.02926460	-0.01898552

- 3) Contingency table

	clust	
quality	1	2
bad wine	725	623
good wine	761	1852





What about the results?

- Negative values for the first component tend to worsen the quality of the wine, positive values lead to good wines,
- High values of residual.sugar, density, total.sulfur.dioxide and alcohol lead to a good wine